mixed with grease, paraffin, kerosene and turpentine. It is applied either on the buffing wheel or on the work. Buffing wheels are rotated against the work to get a superior finish.

Scraping

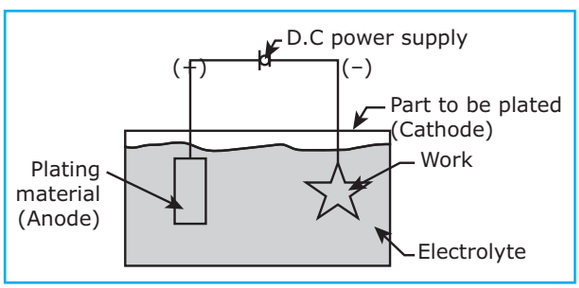
There will be always some minor imperfections on the machined surfaces. They are removed by using hand tool called scraper. There are three types of scrapers-flat, half-round and triangular. The part to be scraped is fitted in a vise and thin layer of Persian blue is applied on the surface. Thin flakes of metal are removed. Persian blue is once again applied to check the flatness.



Scraping

Electro plating

Electro plating is the process of applying metallic coatings on the surfaces of metal parts. It can be done on parts of non-metals also.



Electroplating

This is done for protection against corrosion or against wear and tear and far better appearance. It is also done to slightly increase the size of worn out parts and to make parts easy to solder. It may also used to keep off selected areas on steel parts from being carburized during heat treatment.

Common plating materials are chromium, nickel, copper, zinc, cadmium, etc. The more precious metals like silver, gold, platinum and radium are also applied for plating. Door handles and automobile parts are chromium plated for appearance. The method of plating a layer of zinc is known as galvanizing.

Surface to be plated must be buffed smooth to eliminate scratches. The surface is cleaned by suitable cleaning solutions to remove all grease and dirt.

The four essential elements of a plating process are the cathode, anode, electrolyte and direct current. The current leaves the anode, which is a bar of plating metal and migrates through the electrolyte to the cathode which is the part to be plated.

4.17 Safety precautions

1. We should ensure that the work is held firmly and properly. The grinding wheel should be inspected and mounted on the spindle.
2. Proper work speed, wheel speed and table feed should be selected according to the nature of the work.
3. Safety goggles should be worn by the operator.
4. It should be checked whether the safety guards are fitted.

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